Chapter 1

Introduction

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Research Techniques

1. Research
   - Research Aim and Objectives
   - Features

2. Types of Research
   - Descriptive vs. Analytical Research
   - Applied vs. Fundamental Research
   - Quantitative vs. Quantitative Research
   - Conceptual vs. Empirical

3. 6Ps of Research
   - Purpose of Research
   - Product of Research

Research

- A search for knowledge
- A scientific and systematic search for significant information on a specific topic
- The purpose of research is to answer questions and acquire new knowledge
  - seeks the answer of certain questions which have not been answered so far and the answers depend upon human efforts
- For example: few years ago: did not know exactly the moon's south side?
  - make some assumptions about it
  - by efforts, now able to give concrete answer of the problem

Research

- Research answers only those questions of which the answers are not available in literature i.e., in human knowledge
  - an original contribution to the existing stock of knowledge
- most important process for advancing knowledge to promote progress

MOTIVATION IN RESEARCH

- What makes people undertake research?
  1. Desire to face the challenge of solving the unsolved problems
     - i.e., concern over practical problems initiates research
  2. Desire to get intellectual joy of doing some creative work
  3. Desire to be of service to society

Research Aim

- to improve research procedures through the refinement and extension of knowledge
  - to discover answers to questions through the application of scientific procedures
- each research study has its own specific purpose

Example:

"To investigate the effectiveness of machine learning algorithms in detecting cybersecurity threats in network traffic."

Research Objectives

1. To gain familiarity with a phenomenon or to achieve new insights into it
   - exploratory or formulative studies
2. To portray accurately the characteristics of a particular individual, situation or a group
   - descriptive studies
3. To determine the frequency with which something occurs or with which it is associated with something else
   - diagnostic studies
4. To test a hypothesis of a causal relationship between variables
   - hypothesis-testing studies

Exploratory or Formulative Study

- to gain familiarity with a phenomenon or to achieve new insights into it
- typically conducted when there is limited existing knowledge or understanding of the topic
- lay the groundwork for further research and may help in the formulation of hypotheses or the development of theoretical frameworks

Example

- To review existing literature on machine learning techniques applied to cybersecurity and network traffic analysis

Descriptive Study

- to portray accurately the characteristics of a particular individual, situation, or group
- focus on describing what currently exists or what has occurred, without attempting to determine causality or establish relationships between variables
- to provide a snapshot of a given phenomenon, helping researchers to understand its features, patterns

Example

- To identify common types of cybersecurity threats encountered in network traffic, such as malware, intrusion attempts, and data breaches
- To preprocess the dataset by extracting relevant features and preparing it for training and testing the machine learning models

Diagnostic Study

- To determine the frequency with which something occurs or with which it is associated with something else
- To identify and analyze relationships between variables, uncover patterns and diagnose problems or conditions

Example

- Conducting statistical analyses to determine the correlation between the performance of machine learning algorithms in detecting cybersecurity threats and characteristics of the network traffic data, such as traffic volume, data distribution, and network topology
- Identifying and defining key performance indicators (KPIs) for evaluating the effectiveness of machine learning algorithms in detecting cybersecurity threats, such as detection rate, false positive rate, and response time

Hypothesis testing Study

- To test a hypothesis of a causal relationship between variables
- To determine whether there is a statistically significant relationship between variables
  - ANOVA test
  - t-test

Case Study: Research Objectives

Software engineer wants to develop a e-commerce website. Design a step-by-step to fulfill research objectives

- **Exploratory or Formulative:** Conduct a preliminary investigation into the e-commerce industry, trends, and consumer behavior
  - Explore existing e-commerce websites to gain insights into design trends, features, and user experience
  - Research the technologies and platforms commonly used for developing e-commerce websites
- **Descriptive:** Define the target audience for the e-commerce website, including demographics, preferences, and buying behavior
  - Conduct market research to identify competitors, their products, pricing strategies, and unique selling propositions
  - Document the key features and functionalities required for the e-commerce website based on user needs and market analysis

- **Diagnostic:** Analyze existing e-commerce platforms or websites to identify common issues, challenges, and pain points faced by users
- **Hypothesis-Testing :**Formulate hypotheses about the potential impact of specific variables (e.g., website design, product pricing, marketing strategies) on user engagement and conversion rates
  - Design experiments or A/B tests to systematically test the hypotheses by manipulating variables and measuring outcomes
  - Refine and optimize the e-commerce website based on the findings from hypothesis-testing research to improve its effectiveness in achieving business objectives

Features of Research

1. **Systematic Process**
2. **Objective Orientation**
3. **Empirical Basis**
4. **Logical Reasoning**
5. **Cumulative Nature**
6. **Generalizability**
7. **Ethical Considerations**
8. **Communication and Dissemination**
9. **Continuous Improvement**

Systematic Process

- follows a systematic and structured process that involves:
  - formulation of a problem
    - research questions or hypotheses or problem statement
  - data collection
  - analysis
  - interpretation of findings

Example

- Conducting a study to investigate the impact of mindfulness meditation on stress levels among college students
- The research follows a systematic process
  - defining research questions
  - designing a randomized controlled trial
  - collecting data through surveys and physiological measures
  - analyzing the data using statistical methods
  - interpreting the findings

Objective Orientation

- driven by specific objectives, aims, or questions that guide the inquiry and provide clarity about the desired outcomes

Example

- Examining the relationship between sleep duration and academic performance among students
- Driven by specific objectives to determine if there is a correlation between the two variables (sleep duration and academic performance)

Empirical Basis

- empirical evidence obtained through observation, experimentation, or data collection from the real world
- emphasizes the use of verifiable and replicable methods to support conclusions and findings

Example

- Studying the effectiveness of a new teaching method on student learning outcomes in mathematics
- The research collects quantitative data on student performance scores before and after implementing the teaching method, providing empirical evidence to support conclusions about its efficacy

Logical Reasoning

- involves logical reasoning and critical thinking to analyze data, draw conclusions, and make inferences

Example

- Investigating the impact of environmental factors on public health outcomes, such as air pollution levels on respiratory illnesses
- The research applies
  - logical reasoning to analyze data
  - consider alternative explanations
  - draw conclusions about the causal relationship between environmental exposures and health outcomes

Cumulative Nature

- builds upon existing knowledge and contributes to the collective body of knowledge within a field or discipline

Example

- Examines the effects of rising global temperatures on agricultural productivity across different regions
- Findings contribute to the cumulative body of knowledge on climate change impacts
  - inform strategies for mitigating its effects on food security

Generalizability

- aims to generate findings that are applicable beyond the specific context or sample studied
- establish generalizable principles, patterns, or trends that can be applied to broader populations or situations

Example

- Examining the relationship between sleep duration and academic performance among students
- Findings based on a diverse sample of participants
  - provides insights into generalizable principles to find the relationship between sleep duration and academic performance

Ethical Considerations

- Research adheres to ethical principles and guidelines to ensure
  - the protection of human subjects
  - the responsible conduct of research
  - the integrity of the research process

Example

- Conducting a study on sensitive topics such as mental health or substance abuse
- Researchers obtain informed consent from participants
  - ensures the confidentiality of data
  - provide appropriate support or referrals for individuals experiencing distress

Communication and Dissemination

- Research findings are communicated through scholarly publications, presentations, and other channels to share knowledge, facilitate peer review, and contribute to the advancement of the field

Example

- Publishing research findings on the efficacy of a new teaching method on student learning outcomes in mathematics
- findings reach relevant stakeholders
  - contribute to advancements in teaching area

Continuous Improvement:

- Research is an iterative process that involves ongoing reflection, refinement, and revision
- Researchers continuously seek to improve their methods, theories, and approaches based on feedback, critique, and new insights

Example

- Revising research methods based on feedback from peer reviewers to enhance the validity and reliability of study findings
- Researchers may refine study designs, adjust data collection procedures, or incorporate additional analyses to address limitations and improve the quality of their research

Types of Research

1. **Descriptive vs. Analytical**
2. **Applied vs. Fundamental**
3. **Quantitative vs. Qualitative**
4. **Conceptual vs. Empirical**

Descriptive Research

- To portray accurately the characteristics of a particular individual, situation, or group
- focuses on describing what currently exists or what has occurred, without attempting to determine causality or establish relationships between variables

Methodology

- observational methods, surveys, questionnaires, and other data collection techniques to gather information about the subject of study
- emphasizes the collection of quantitative data to describe and summarize the features, behaviours, or attributes of interest

Analytical Research

- researcher has to use facts or information already available, and analyze these to make a critical evaluation of the material
- interpret data to uncover patterns, relationships, or causal links between variables
- seeks to understand why certain phenomena occur, predict future outcomes, or test hypotheses about the relationships between variables

Methodology

- employs statistical analysis, experimentation, modeling, and other advanced techniques to examine the relationships between variables and draw conclusions based on empirical evidence
- involves comparing and contrasting data, identifying trends, and testing hypotheses to infer causal relationships

Applied Research

- To find a solution for an immediate or practical problem facing a society or an industrial/business organisation
- Researchers collaborate with industry or government partners to tackle specific problems
- The results of applied research often lead to the development of new products, technologies, or processes

Fundamental Research

- also known as basic research, aims to deepen our understanding of fundamental principles, theories, and concepts
- driven by curiosity and seeks to explore the underlying mechanisms without immediate practical applications
- Researchers explore mathematical models, proofs, and abstract concepts

Quantitative Research

- involves collecting and analyzing numerical data to describe, predict, or control variables of interest
- aims to test causal relationships between variables, make predictions, and generalize results to wider populations

Qualitative Research

- understand the underlying meanings, experiences, and social contexts related to a phenomenon relating to or involving quality
- seeks to explore subjective experiences, beliefs, and behaviors

Conceptual Research

- also known as theoretical research, focuses on developing theories, models, or frameworks
- aims to deepen our understanding of fundamental principles and concepts
    - contributes to the theoretical foundation of a field

Empirical Research

- involves collecting and analyzing data to answer specific research questions
  - data-based research
- come up with conclusions which are capable of being verified by observation or experiment

6Ps of Research

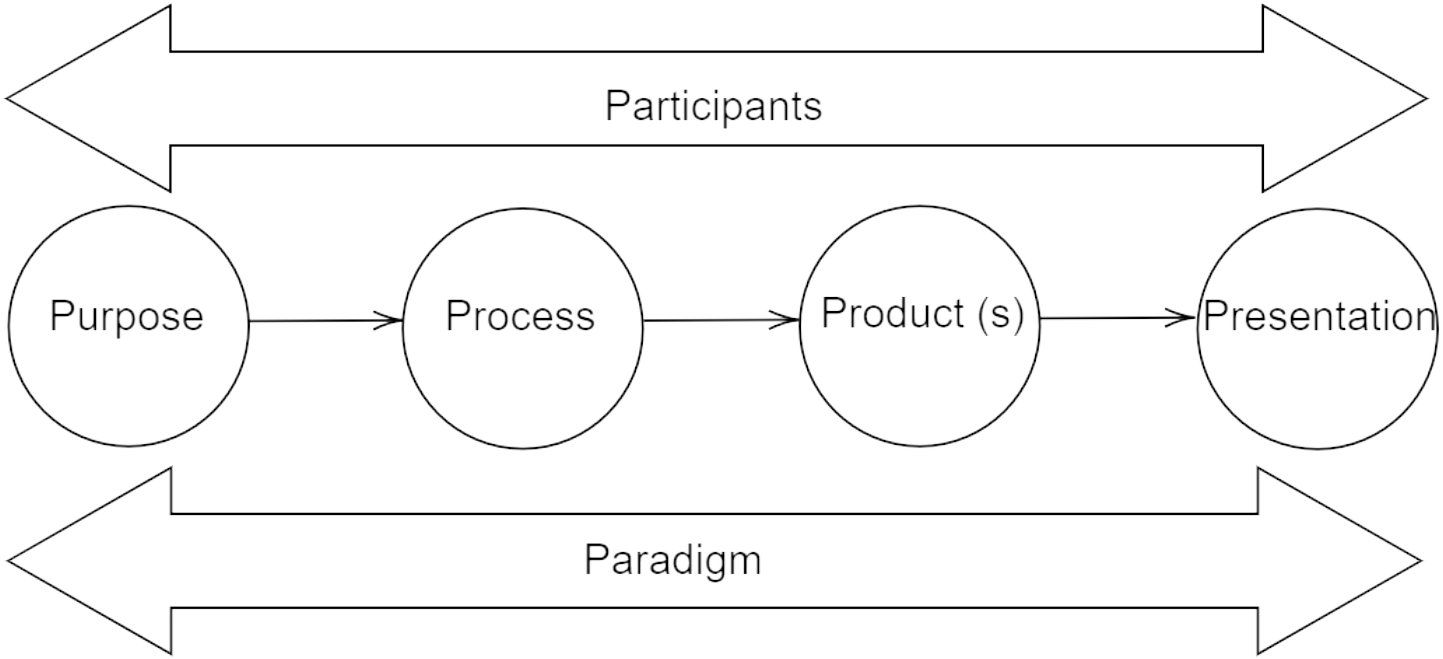
1. **Purpose**
2. **Product**
3. **Process**
4. **Participants**
5. **Paradigm**
6. **Presentation**

6Ps of Research

1. **Purpose:** the reason for conducting research
   - It clarifies what you aim to achieve and guides your research questions
   - why it is important or useful to study this, the specific research question(s) asked and the objectives set
   - Research without a purpose is unlikely to be good research
2. **Product:** outcomes of research, especially your contribution to knowledge about your subject area
   - include research papers, software prototypes, algorithms
   - Think about what you intend to produce as a result of your research effort
3. **Process:** the sequence of activities undertaken in any research
   - involves identifying research topics, establishing a conceptual framework (how you choose to think about your research topic)
   - selecting and using a research strategy and data generation methods, analyzing data
   - drawing conclusions, and recognizing any limitations in your research

4. **Participants:** individuals or groups involved in research
   - directly involvement: by interviewing them or observing them
   - indirectly involvement: editors to whom you submit a research paper
5. **Paradigm:** a pattern or model or shared way of thinking
   - that shapes research approach
   - need for a 'paradigm shift' to mean that a new way of thinking is required
6. **Presentation:** means by which the research is disseminated and explained to others
   - written up in a paper or thesis, or a conference paper is presented to an audience of conference delegates
   - a computer-based product is demonstrated to clients, users or examiners
   - Important that the presentation is carried out professionally
     - otherwise your audience might assume your whole research project was not undertaken in a professional manner

6Ps of Research



Purpose of Research

- incorporates the reason for doing it, the topic of interest
- why it is important or useful to study this
- the specific problem statement or research question(s) asked and the objectives set
- Why do people do research?
  - To add to the body of knowledge
  - To solve a problem
  - To find out what happens
  - To find the evidence to inform practice
  - To develop a greater understanding of people and their world

Product of Research

- A new or improved product
- A new theory
- A re-interpretation of an existing theory